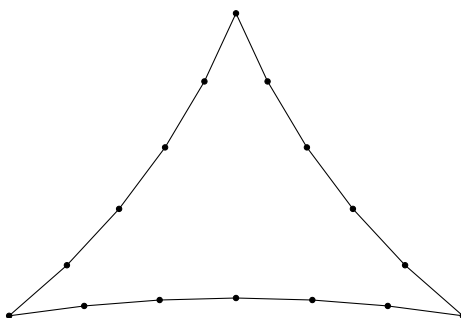


Problem 37. Let $n \geq 3$ be an integer. What is the largest possible number of interior angles greater than 180° in an n -gon in the plane, given that the n -gon does not intersect itself and all its sides have the same length?

Answer. 0 if $n = 3, 4$ and $n - 3$ for $n \geq 5$.

Solution. If $n = 3, 4$ then any such n -gon is a triangle, respectively a rhombus, therefore the answer is 0. If $n = 5$ then consider a triangle with side lengths 2, 2, 1. Now move the vertex between sides of length 2 towards the opposite side by 0.0000001 units. Consider the triangle as a closed physical chain of links of length 1 that are aluminium tubes and through them is a closed rubber string. So deform the chain on a level surface so that links of the chain move towards the inside of the triangle, by fixing the vertices of the triangle to the surface. Geometrically, the links which are on sides of length 2 are now distinct chords of a large circle, attached to each other by their endpoints. If $n > 5$ then first consider a triangle of integer sides lengths, with sides of as equal lengths as possible, so that the sum of side lengths is n . Imagine a closed aluminium chain on a rubber string, as in the previous case. Now move two of the vertices of the triangle towards the third one by 0.00000001 units each. Again consider the chain on a level surface and deform it so that its links move towards the interior of the triangle. Geometrically each side of a triangle is deformed into consecutive equal-length chords of a large circle with centre far away from the original triangle. For $n \geq 5$ this gives an example of $n - 3$ interior angles of more than 180° . To see that $n - 2$ or more such angles is not possible, note that the sum of interior angles of any n -gon (that does not cut itself) is equal to $(n - 2)180^\circ$, but already the sum of the 'large' angles would be greater than that if there were at least $n - 2$ of 'large' angles of size greater than 180° .



□

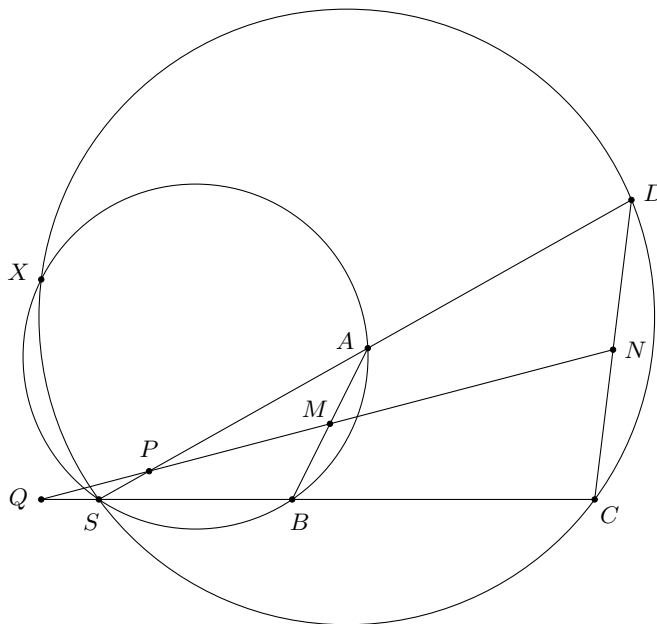
Problem 38. In a country there are 2018 airports, no three of them lying on a line. Two airports are connected by a direct flight if and only if the line passing through them divides the country in two parts, each with 1008 airports in it. Show that there are no two airports such that one can travel from the first to the second, visiting each of the 2018 airports exactly once.

Solution. Denote airports as points on the plane. Each airport that is a vertex of the convex hull of these points has only one direct flight. (When we rotate the line that passes through such point the numbers of other points in the half-planes change monotonically.) If an airport has only one direct flight then it can be only the starting point or the endpoint of the journey that visits all 2018 airports. The convex hull contains at least 3 vertices, so there are at least three airports that has only one direct flight. It means that such a journey is impossible. □

Problem 39. Let $ABCD$ be a convex quadrilateral such that $BC = AD$. Let M and N be the midpoints of AB and CD , respectively. The lines AD and BC meet the line MN at P and Q , respectively. Prove that $CQ = DP$.

Solution. Let AD meet BC at S . Let X be the intersection of the circumcircles of $\triangle ABS$ and $\triangle CDS$. Then X is the centre of spiral similarity mapping AB to DC . Thus, $\triangle XAD \sim \triangle XBC$. Since $AD = BC$, they are indeed congruent, and hence $XD = XC$. As M and N are the midpoints of AB and DC respectively, the above spiral similarity also maps M to N .

As X is the centre of spiral similarity mapping AM to DN , points X, P, N, D are concyclic. Similarly, X, Q, C, N are concyclic. Hence, X is the centre of spiral similarity mapping QP to CD . Thus, it is also the centre of spiral similarity mapping PD to QC . This shows $\triangle XPD \sim \triangle XQC$. As $XD = XC$, the triangles are congruent. This implies $PD = QC$.



□

Problem 40. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ for which

$$f(x+y) = f(x-y) + f(f(1-xy))$$

holds for all real numbers x and y .

Answer. $f(x) \equiv 0$.

Solution. Substituting $y = 0$ gives $f(x) = f(x) + f(f(1))$, hence $f(f(1)) = 0$. Using this after substituting $x = 0$ into the original equation gives $f(y) = f(-y)$ for all y , i.e., f is even. Substituting $x = 1$ into the original equation gives $f(1+y) = f(1-y) + f(f(1-y))$. Since f is even, we have $f(-1-y) = f(1-y) + f(f(1-y))$. Hence $f(f(1-y)) = f(1-y-2) - f(1-y)$. As $1-y$ covers all real values, one can conclude that

$$f(f(z)) = f(z-2) - f(z) \quad (1)$$

for all real numbers z . Substituting $-z$ for z into (1) and simplifying the terms by using the fact that f is even, one obtains $f(f(z)) = f(z+2) - f(z)$. Together with (1), this implies

$$f(z+2) = f(z-2) \quad (2)$$

for all real numbers z . Now taking $y = 2$ in the original equation followed by applying (2) leads to $f(f(1-2x)) = 0$ for all real x . As $1-2x$ covers all real values, one can conclude that

$$f(f(z)) = 0 \quad (3)$$

for all real numbers z . Thus the original equation reduces to $f(x+y) = f(x-y)$. Taking $x = y$ here gives $f(2x) = f(0)$, i.e., f is constant, as $2x$ covers all real numbers. As 0 must be among the values of f by (3), $f(x) \equiv 0$ is the only possibility. □